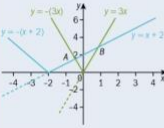


1. Algebra Vault — Algebra

a. Absolute value/modulus function -> graph + solving equations

Example 1
Solve the equation $|x + 2| = |3x|$.

Method 1: Using a graph



Sketch the graphs and find where they intersect.
The lines cannot be drawn below the x -axis.
For $x < 0$, $|3x| = -(3x)$
For $x < -2$, $|x + 2| = -(x + 2)$
Graphs intersect at A and B.

At A, $x + 2 = -3x$
 $4x = -2 \Rightarrow x = -\frac{1}{2}$

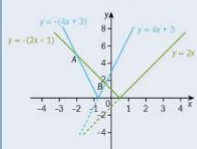
At B, $x + 2 = 3x$
 $2x = 2 \Rightarrow x = 1$
 $x = -\frac{1}{2}$ or $x = 1$

At A, the line $y = x + 2$ intersects the line $y = -(3x)$.

At B, the line $y = x + 2$ intersects the line $y = 3x$.

Example 2
Solve the inequality $|4x + 3| > |2x - 1|$.

Method 1: Using a graph



Sketch the graphs and find where they intersect.
The lines cannot be drawn below the x -axis.
For $x < -\frac{3}{4}$, $|4x + 3| = -(4x + 3)$
For $x < \frac{1}{2}$, $|2x - 1| = -(2x - 1)$
Graphs meet at A and B.

At A, the line $y = -(4x + 3)$ intersects the line $y = -(2x - 1)$.

At B, the line $y = 4x + 3$ intersects the line $y = -(2x - 1)$.

At A, $-(4x + 3) = -(2x - 1)$
 $2x = -4$, $x = -2$

At B, $4x + 3 = -(2x - 1)$
 $6x = -2$, $x = -\frac{1}{3}$

We want the region where $|4x + 3| > |2x - 1|$.
This is where the 'blue' lines are above the 'green' lines.
 $x < -2$ and $x > -\frac{1}{3}$

1. Solve each of these equations algebraically.

- | | |
|--------------------------|---------------------------|
| a) $ 1 - 2x = 3$ | b) $ x - 3 = x + 1 $ |
| c) $ 5x - 2 = 2x $ | d) $ 5 - 4x = 4$ |
| e) $ 2x - 1 = x + 2 $ | f) $ x = 4 - 2x $ |
| g) $ 3x + 1 = 4 - 2x $ | h) $ 2x - 6 = 3x + 1 $ |
| i) $ x + 4 = 3x + 1 $ | j) $ 1 - 3x = 5x - 3 $ |
| k) $3 x - 4 = x + 2 $ | l) $5 2x - 3 = 4 x - 5 $ |

2. Solve each of these inequalities algebraically.

- | | |
|----------------------------|-----------------------|
| a) $ 2x - 3 < x $ | b) $ x - 1 \geq 4$ |
| c) $ x + 3 \geq 2x + 2 $ | d) $ 2x + 3 > x + 6$ |

3. Solve each of these inequalities graphically.

- | | |
|----------------------------|-----------------------------|
| a) $ x + 6 \leq 3 x - 2 $ | b) $ 3x - 2 < x + 4 $ |
| c) $ 2x < 1 - x $ | d) $5 \leq 2x - 1 $ |
| e) $2 x - 1 < x + 3 $ | f) $ 2x + 1 \geq 1 - 4x $ |
| g) $ x + 2 < 2 x + 1 $ | h) $ 3x - 1 \leq x + 3 $ |

b. Division of polynomials

Example 4

Divide $x^3 - 5x^2 + x + 10$ by $(x - 2)$.

$\begin{array}{r} x^2 - 3x - 5 \\ x - 2 \overline{) x^3 - 5x^2 + x + 10} \\ \underline{x^3 - 2x^2} \\ -3x^2 + x + 10 \\ \underline{-3x^2 + 6x} \\ -5x + 10 \\ \underline{-5x + 10} \\ 0 \end{array}$	$x^3 \div x = x^2$, so multiply $(x - 2)$ by x^2.
	Subtract $(x^3 - 2x^2)$ from $(x^3 - 5x^2)$ and bring down $(+x + 10)$.
	$-3x^2 \div x = -3x$, so multiply $(x - 2)$ by $-3x$.
	Subtract $(3x^2 + 6x)$ from $(3x^2 + x)$ and bring down $+10$.

We cannot continue the process because $(-5x + 10) - (-5x + 10) = 0$.

Thus $(x^3 - 5x^2 + x + 10) \div (x - 2) = x^2 - 3x - 5$

Example 5

Find the remainder when $4x^3 - 7x - 1$ is divided by $(2x + 1)$.

$\begin{array}{r} 2x^2 - x - 3 \\ 2x + 1 \overline{) 4x^3 + 0x^2 - 7x - 1} \\ \underline{4x^3 + 2x^2} \\ -2x^2 - 7x - 1 \\ \underline{-2x^2 - x} \\ -6x - 1 \\ \underline{-6x - 3} \\ +2 \end{array}$	As there is no term in x^2 in the dividend, it is useful to write $0x^2$ as part of the dividend.
	$(4x^3 \div 2x) = 2x^2$, so multiply $(2x + 1)$ by $2x^2$.
	Subtract $(4x^3 + 2x^2)$ from $(4x^3 + 0x^2)$ and bring down $-7x - 1$.
	$(-2x^2 \div 2x) = -x$, so multiply $(2x + 1)$ by $-x$.
	Subtract $(-2x^2 - x)$ from $(-2x^2 - 7x)$ and bring down -1.
	Subtract $(-6x - 3)$ from $(-6x - 1)$ to get a remainder of $+2$.

We cannot continue the process as 2 cannot be divided by $(2x + 1)$.

Thus, when $(4x^3 - 7x - 1)$ is divided by $(2x + 1)$, the remainder is 2.

1. Divide

- $x^3 + 3x^2 + 3x + 2$ by $(x + 2)$
- $x^3 - 2x^2 + 6x + 9$ by $(x + 1)$
- $x^3 - 3x^2 + 6x - 8$ by $(x - 2)$
- $x^3 + x^2 - 3x - 2$ by $(x + 2)$
- $2x^3 - 6x^2 + 7x - 21$ by $(x - 3)$
- $3x^3 - 20x^2 + 10x + 12$ by $(x - 6)$
- $6x^4 + 5x^3 + 5x^2 + 10x + 7$ by $(3x^2 - 2x + 4)$.

Hint: In part (g), use the same method as when dividing by a linear expression. State any remainder.

2. Find the remainder when

- $6x^3 + 28x^2 - 7x + 10$ is divided by $(x + 5)$
- $2x^3 + x^2 + 5x - 4$ is divided by $(x - 1)$
- $x^3 + 2x^2 - 17x - 2$ is divided by $(x - 3)$
- $2x^3 + 3x^2 - 4x + 5$ is divided by $(x^2 + 2)$
- $4x^3 - 5x + 4$ is divided by $(2x - 1)$
- $3x^3 - x^2 + 1$ is divided by $(x + 2)$.

3. **Show that** $(2x + 1)$ is a factor of $2x^3 - 3x^2 + 2x + 2$.

4. a) Show that $(x - 1)$ is a factor of $x^3 - 6x^2 + 11x - 6$.

b) Hence factorise $x^3 - 6x^2 + 11x - 6$.

5. Show that when $4x^3 - 6x^2 + 5$ is divided by $(2x - 1)$ the remainder is 4.

6. Divide $x^3 + 1$ by $(x + 1)$.

7. Find the quotient and the remainder when $x^4 + 2x^3 + 3x^2 + 7$ is divided by $(x^2 + x + 1)$.

8. Find the quotient and the remainder when $2x^3 + 3x^2 - 4x + 5$ is divided by $(x + 2)$.

c. Remainder/factor theorem

Example 6

Find the remainder when $4x^3 + x^2 - 3x + 7$ is divided by $(x + 2)$.

$$f(x) = 4x^3 + x^2 - 3x + 7$$

Write the polynomial as a function.

$$\begin{aligned} f(-2) &= 4(-2)^3 + (-2)^2 - 3(-2) + 7 \\ &= -32 + 4 + 6 + 7 = -15 \end{aligned}$$

$(x + 2) = 0 \Rightarrow x = -2$, so calculate $f(-2)$.

The remainder is -15 .

Example 7

When $16x^4 - ax^3 + 8x^2 - 4x - 1$ is divided by $(2x - 1)$, the remainder is 3.

Find the value of a .

$$f(x) = 16x^4 - ax^3 + 8x^2 - 4x - 1$$

Write the polynomial as a function.

$$f\left(\frac{1}{2}\right) = 16\left(\frac{1}{2}\right)^4 - a\left(\frac{1}{2}\right)^3 + 8\left(\frac{1}{2}\right)^2 - 4\left(\frac{1}{2}\right) - 1$$

$(2x - 1) = 0 \Rightarrow x = \frac{1}{2}$, so calculate $f\left(\frac{1}{2}\right)$.

$$= 1 - \frac{1}{8}a + 2 - 2 - 1 = 3$$

Equate this to 3, since the remainder is 3.

$$-\frac{1}{8}a = 3 \Rightarrow a = -24$$

Example 9

Solve $x^3 - 3x^2 - 4x + 12 = 0$.

Let $f(x) = x^3 - 3x^2 - 4x + 12$.

$$f(1) = 1 - 3 - 4 + 12 \neq 0$$

To solve, we must first factorise $x^3 - 3x^2 - 4x + 12$.

so $(x - 1)$ is not a factor.

Trial any value of x that is a factor of 12.

$$f(2) = 8 - 12 - 8 + 12 = 0$$

so $(x - 2)$ is a factor.

Alternatively, at this stage you could also do a long division to find the other two factors since you already know one factor.

$$f(-2) = -8 - 12 + 8 + 12 = 0$$

so $(x + 2)$ is a factor.

We can deduce that the third factor is $(x - 3)$.

$12 \div 2 \div -2 = -3$ (and the coefficient of x^3 is 1).

$$f(3) = 27 - 27 - 12 + 12$$

so $(x - 3)$ is a factor.

Thus $(x - 2)(x + 2)(x - 3) = 0$

$$x = 2 \quad \text{or} \quad x = -2 \quad \text{or} \quad x = 3$$

Example 10

The polynomial $ax^3 + x^2 + bx + 6$, where a and b are constants, is denoted by $p(x)$. It is given that $(2x - 1)$ is a factor of $p(x)$ and that when $p(x)$ is divided by $(x - 1)$ the remainder is -4 .

Find the values of a and b .

$$p(x) = ax^3 + x^2 + bx + 6$$

$$p\left(\frac{1}{2}\right) = \frac{a}{8} + \frac{1}{4} + \frac{b}{2} + 6 = 0$$

$p\left(\frac{1}{2}\right) = 0$ as $(2x - 1)$ is a factor.

$$a + 2 + 4b + 48 = 0$$

Multiply each term by 8.

$$a + 4b = -50 \quad (1)$$

$$p(1) = a + 1 + b + 6 = -4$$

$p(1) = -4$ as the remainder is -4 .

$$a + b = -11 \quad (2)$$

$$(1) - (2) \Rightarrow 3b = -39$$

Solve the two equations simultaneously.

$$b = -13$$

$$(2) \Rightarrow a - 13 = -11$$

$$a = 2$$

$$a = 2 \text{ and } b = -13$$

1. Find the remainder when
 - a) $2x^3 + 8x^2 - x + 4$ is divided by $(x - 3)$
 - b) $5x^4 - 3x^3 - 2x^2 + x - 1$ is divided by $(x + 1)$
 - c) $x^3 + 4x^2 + 8x - 3$ is divided by $(2x + 1)$
 - d) $3x^3 - 2x^2 - 5x - 7$ is divided by $(2 - x)$
 - e) $9x^3 - 8x + 3$ is divided by $(1 - x)$
 - f) $243x^4 - 27x^3 + 6x + 4$ is divided by $(3x - 2)$.
2. When $ax^3 + 16x^2 - 5x - 5$ is divided by $(2x - 1)$ the remainder is -2 . Find the value of a .
3. The polynomial $4x^3 - 4x^2 + ax + 1$, where a is a constant, is denoted by $p(x)$. When $p(x)$ is divided by $(2x - 3)$ the remainder is 13. Find the value of a .
4. The polynomial $x^3 + ax^2 + bx + 1$, where a and b are constants, is denoted by $p(x)$. When $p(x)$ is divided by $(x - 2)$ the remainder is 9 and when $p(x)$ is divided by $(x + 3)$ the remainder is 19. Find the value of a and the value of b .
5. When $5x^3 + ax + b$ is divided by $(x - 2)$, the remainder is equal to the remainder obtained when the same expression is divided by $(x + 2)$.
5. $(x - 2)$ is a factor of $x^3 - 3x^2 + ax - 10$. Evaluate the coefficient a .
6. a) Show that $(2x - 5)$ is a factor of $4x^3 - 20x^2 + 19x + 15$.
b) Hence factorise $4x^3 - 20x^2 + 19x + 15$ as a product of three linear factors.
7. The polynomial $ax^3 - 3x^2 - 5ax - 9$ is denoted by $p(x)$ where a is a real number. It is given that $(x - a)$ is a factor of $p(x)$. Find the possible values of a .
8. The polynomial $3x^3 + 2x^2 - bx + a$, where a and b are constants, is denoted by $p(x)$. It is given that $(x - 1)$ is a factor of $p(x)$ and that when $p(x)$ is divided by $(x + 1)$ the remainder is 10. Find the values of a and b .
9. The polynomial $ax^3 + bx^2 - 5x + 3$, where a and b are constants, is denoted by $p(x)$. It is given that $(2x - 1)$ is a factor of $p(x)$ and that when $p(x)$ is divided by $(x - 1)$ the remainder is -3 . Find the remainder when $p(x)$ is divided by $(x + 3)$.
10. Factorise $2x^4 + 5x^3 - 5x - 2$ as a product of four linear factors.

d. Partial fractions (3 cases)

Partial fractions type 1: Algebraic fractions with linear factors in the denominator can be split into partial fractions in the following way:

$$\frac{1}{(x+p)(x+q)} \equiv \frac{A}{(x+p)} + \frac{B}{(x+q)}$$

Example 1

Express $\frac{x+3}{(x-2)(x+1)}$ in partial fractions.

$$\frac{x+3}{(x-2)(x+1)} \equiv \frac{A}{x-2} + \frac{B}{x+1}$$

The denominator has two linear factors so we use two fractions.

$$x+3 \equiv A(x+1) + B(x-2)$$

Multiply each term by $(x-2)(x+1)$.

Method 1: Choose appropriate values for x .

We use the symbol $\equiv$ to mean it is true for all values of x .

Let $x = 2$

Choose a value of x that makes one bracket = 0.

$$5 = A(3) + 0 \Rightarrow A = \frac{5}{3}$$

Let $x = -1$

Choose a value of x that makes the other bracket = 0.

$$2 = 0 + B(-3) \Rightarrow B = -\frac{2}{3}$$

$$\text{Hence } \frac{x+3}{(x-2)(x+1)} \equiv \frac{5}{3(x-2)} - \frac{2}{3(x+1)}$$

Substitute the values of A and B into the fractions.

Method 2: Expand the brackets and equate coefficients.

$$x+3 \equiv A(x+1) + B(x-2)$$

$$x+3 \equiv Ax + A + Bx - 2B$$

Expand the brackets.

$$(1)x + (3) \equiv (A+B)x + (A-2B)$$

$$\text{Hence } A+B=1 \text{ and } A-2B=3$$

Equate the coefficients of x and equate the constants.

Solving simultaneously gives $A = \frac{5}{3}$ and $B = -\frac{2}{3}$

$$\text{Hence } \frac{x+3}{(x-2)(x+1)} \equiv \frac{5}{3(x-2)} - \frac{2}{3(x+1)}$$

Put the minus sign (-) before the second fraction.

Partial fractions type 2: Algebraic fractions with a repeated factor in the denominator can be split into partial fractions in the following way:

$$\frac{1}{(x+p)^2} \equiv \frac{A}{(x+p)} + \frac{B}{(x+p)^2}$$

Example 3

Express $\frac{3x^2+2}{x(x-1)^2}$ in partial fractions.

$$\frac{3x^2+2}{x(x-1)^2} \equiv \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

Separate the repeated factor $(x-1)^2$ into two denominators.

$$3x^2+2 \equiv A(x-1)^2 + Bx(x-1) + Cx$$

Let $x = 0$

This means B and C are eliminated.

$$2 = A(-1)^2 + 0 + 0 \Rightarrow A = 2$$

Let $x = 1$

$$5 = 0 + 0 + C \Rightarrow C = 5$$

This means A and B are eliminated.

Equating coefficients of x^2 :

$$3 = A + B \Rightarrow 3 = 2 + B \Rightarrow B = 1$$

Substitute for $A = 2$ in $3 = A + B$.

$$\frac{3x^2+2}{x(x-1)^2} \equiv \frac{2}{x} + \frac{1}{x-1} + \frac{5}{(x-1)^2}$$

Note: You could also equate coefficients of x or equate the constant terms.

Partial fractions type 3: Algebraic fractions with a quadratic factor in the denominator can be split into partial fractions in the following way:

$$\frac{1}{(x+p)(x^2+q)} \equiv \frac{A}{x+p} + \frac{Bx+C}{x^2+q}$$

Example 5

Express $\frac{2x^2 - x + 6}{(x+1)(x^2+2)}$ in partial fractions.

$$\frac{2x^2 - x + 6}{(x+1)(x^2+2)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+2}$$

Put $Bx + C$ as the numerator of $x^2 + 2$.

$$2x^2 - x + 6 \equiv A(x^2 + 2) + (Bx + C)(x + 1)$$

Let $x = -1$

This means B and C are eliminated.

$$9 = A(3) + 0 \Rightarrow A = 3$$

Equating coefficients of x^2 :

$$2 = A + B \Rightarrow 2 = 3 + B$$

Substitute for $A = 3$ in $2 = A + B$.

$$\Rightarrow B = -1$$

Equating the constant terms:

$$6 = 2A + C \Rightarrow 6 = 6 + C$$

Substitute for $A = 3$ in $6 = 2A + C$.

$$\Rightarrow C = 0$$

$$\frac{2x^2 - x + 6}{(x+1)(x^2+2)} \equiv \frac{3}{x+1} - \frac{x}{x^2+2}$$

Note: You could also equate coefficients of x , but it is simpler in this case to equate coefficients of x^2 and equate the constant terms.

1. Express in partial fractions $\frac{3x+11}{x^2-x-6}$.

2. Express in partial fractions $\frac{2x-1}{(x+1)(x^2+1)}$.

3. Express in partial fractions $\frac{x^2+3x-1}{(x-2)(x+1)}$.

4. Express in partial fractions $\frac{3x+1}{(x-1)^2(x+2)}$.

5. Express in partial fractions

$$\frac{32+45x-8x^2}{10(x+3)(x-2)(2x-1)}$$

6. Express in partial fractions $\frac{96+12x-5x^2}{(x+4)^2(x-4)}$.

7. Express in partial fractions $\frac{7x^2+8x-11}{(x^2-2)(x+3)}$.

e. Binomial expansion (2 cases)

Example 9

Expand $(1-2x)^{-2}$ in ascending powers of x , up to and including the term in x^3 , simplifying the coefficients.

$$(1-2x)^{-2} = 1 + (-2)(-2x) + \frac{(-2)(-3)}{2 \times 1} (-2x)^2 + \frac{(-2)(-3)(-4)}{3 \times 2 \times 1} (-2x)^3 + \dots$$

$n = -2$, and we use $-2x$ instead of x .

$$= 1 + 4x + 12x^2 + 32x^3 + \dots$$

valid for $|-2x| < 1$

Instead of $|x| < 1$ we write $|-2x| < 1$.

i.e. valid for $-\frac{1}{2} < x < \frac{1}{2}$

Write this condition in the form shown.

Example 11

Find the coefficient of x^3 in the expansion of $\frac{3+x}{1+3x}$.

$$\frac{3+x}{1+3x} = (3+x)(1+3x)^{-1}$$

The denominator is the same as $(1+3x)^{-1}$.

$$(1+3x)^{-1} = 1 + (-1)(3x) + \frac{(-1)(-2)}{2 \times 1} (3x)^2 + \frac{(-1)(-2)(-3)}{3 \times 2 \times 1} (3x)^3 + \dots$$

$$= 1 - 3x + 9x^2 - 27x^3 + \dots$$

Expand the denominator.

$$(3+x)(1+3x)^{-1} = (3+x)(1-3x+9x^2-27x^3+\dots)$$

$$\text{Term in } x^3 \text{ is } (3)(-27x^3) + (x)(9x^2) = -72x^3$$

You do not need to fully expand the brackets.

The coefficient of x^3 is -72 .

Do not put x^3 in your answer.

Example 13

Express the following in the form $a^n(1+x)^n$.

a) $(5-x)^{-3}$

b) $\sqrt{(2-6x)}$

c) $\frac{32}{(4+2x)^2}$

$$\text{a) } (5-x)^{-3} = \left[5\left(1 - \frac{x}{5}\right)\right]^{-3}$$

Take out 5 and divide each term in the bracket by 5.

$$= 5^{-3} \left(1 - \frac{x}{5}\right)^{-3}$$

Remember 5 is also to the power of -3 .

$$= \frac{1}{125} \left(1 - \frac{x}{5}\right)^{-3}$$

$$\text{b) } \sqrt{(2-6x)} = [2(1-3x)]^{\frac{1}{2}}$$

It is helpful to use square brackets and round brackets.

$$= 2^{\frac{1}{2}} (1-3x)^{\frac{1}{2}}$$

$$\text{c) } \frac{32}{(4+2x)^2} = 32(4+2x)^{-2}$$

First get rid of the fraction.

$$= 32 \left[4\left(1 + \frac{1}{2}x\right)\right]^{-2}$$

$$= 32 \left[4^{-2} \left(1 + \frac{1}{2}x\right)^{-2}\right]$$

$$= 2 \left(1 + \frac{1}{2}x\right)^{-2}$$

$$4^{-2} = \frac{1}{16}$$

12. a) Expand $\frac{1}{\sqrt{4+2x}}$ in ascending powers of x , up to and including the term in x^3 .
- b) Use your expansion to estimate the value of $4.02^{-\frac{1}{2}}$.
13. Expand $(2-x)(\sqrt{1-2x})$ in ascending powers of x , up to and including the term in x^3 , simplifying the coefficients.
14. a) Express $\frac{2x+5}{(1+x)(2+x)}$ in partial fractions.
- b) Hence obtain the expansion of $\frac{2x+5}{(1+x)(2+x)}$ in ascending powers of x , up to and including the term in x^3 , simplifying your answer.
15. $f(x) = \frac{3x^2 - x + 4}{(1+x)(1-x)^2}$
- a) Express $f(x)$ in partial fractions.
- b) Hence obtain the expansion of $f(x)$ in ascending powers of x , up to and including the term in x^2 .